

What Does a Sim-to-Real Correlation Support? Five One-Line Checks and a Twenty-Two-Paper Audit

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Abstract

Real-to-sim policy evaluation reports its central claim as a correlation between simulated and real-world policy success rates. We survey the twenty-two papers reporting such a correlation and find that the field’s central number rests on almost no independent data. Twenty-one of twenty-two compute their headline correlation over fewer than ten independent training units; five compute it over checkpoints, finetunes, or test conditions of a single policy (one unit); five report any uncertainty on it. Three print $p < 0.001$, parametric values computed over pooled, non-independent points, where their own unit counts ($k = 3$, and in one case $k = 2$) cap any permutation test over units at $p = 0.167$ or 0.5 . These are combinatorial ceilings, not data-availability problems: they would hold if every paper released everything tomorrow.

We propose five checks that should accompany any such correlation, each costing a line of arithmetic on data the authors already have, and we ship them as a tool. Two of the five need only the number of independent units, recoverable from every paper’s own setup description (§8.1 states the coding rules where it is ambiguous). Justifying the rest against data recovered from published figures, we find the conventional signals rank results in exactly the wrong order. One of the survey’s largest designs (RoboWorld, $r = 0.989$, $p < 0.001$) drops to 0.798 when one of its eight policies is removed, and the p -value neither warns nor helps (its rank correlation survives the same removal; §4 reports both); a paper with half the units and no p -value (Digital Cousins) is stable to every removal. That contrast is the good news: the checks separate results that survive scrutiny from results that do not, and they clear more often than they flag. An unstated checkpoint-selection choice flips one published correlation from $+0.90$ to -0.31 ; the ranking metric reported alongside r is roughly three to fifteen times less stable than r itself; one published value of that metric is not attainable at the sample size its own paper states. The same checks, applied to our own pipeline, caught two silently dropped preregistration requirements (§9).

None of this requires new experiments. Five papers already report some uncertainty at these sample sizes. What we are asking for is one parenthetical in a results table:

$$r = 0.989 \text{ (} k = 8 \text{ policies; drop-one range } 0.798\text{--}0.994; \text{ Fisher-}z \text{ [} 0.937, 0.998 \text{])}$$

We release the tool, the recovered data, the analysis, and its preregistration.

1. The setting

Testing a robot policy on hardware is slow and expensive. So the field builds simulators and asks whether policy *ranking* in simulation matches ranking in reality. The answer is reported as a Pearson correlation, and high values are read as license to substitute simulation for hardware.

The design underneath is small. Each correlation is typically computed over a handful of policies, sometimes over multiple training checkpoints of the *same* policy, which are not independent samples. Each point’s success rate is a binomial proportion over 16–40 real episodes.

This is not a criticism of any author: real-robot episodes are genuinely expensive. It does mean the published number carries less information than three decimal places imply. We quantify how much less.

Two findings need stating before any method, because they require no extraction and no judgment call, only counting against each paper’s own setup description. Five of the twenty-two surveyed papers compute their headline correlation over checkpoints or finetunes of a single policy: one independent unit. Three papers print $p < 0.001$ for correlations whose unit counts cap any permutation test over units at $p = 0.167$ or, at $k = 2$, 0.5 . The printed values are parametric

tests over pooled, non-independent points. Nothing was miscalculated; the convention answers a different question from the one the number appears to answer (§8).

The rest of the findings required recovering data from published figures. What checking each named number revealed (MMRV, defined in §7, is the field’s ranking metric: how badly sim mis-ranks the policies, weighted by how much the mistake matters):

published number	what checking revealed
RoboWorld $r = 0.989$, $p < 0.001$	0.798 after removing one of eight policies; the p -value does not warn (§4)
real2sim-eval rope $r = +0.90$	−0.31 under an equally defensible checkpoint-selection rule, which the paper does not state (§5)
real2sim-eval T-block MMRV = 0.108	misses the lattice at the only printed checkpoint count (an appendix count stated for a different experiment); lands on it at the count our extraction recovers (§7.1)
real2sim-eval Table I MMRVs	reproduce to print precision — but only under the reverse of the metric’s defining argument order, identifiable solely by brute-forcing conventions against two figures at once (§7.2)
SimFoundry MMRV = 0.018	our own first verdict against it, withdrawn: the granularity test passes arbitrary values 52.5% of the time there, and is inapplicable to a mean over per-task lattices (§7.1)
MolmoSpaces pick $\rho = 0.98$	appears nowhere in its figures, which print 1.00; our extraction confirms perfectly concordant ranks (§8.1)
REALM V-VIEW $r = 0.89$, MMRV = 0.253	the r reproduces from the panel’s 14 plotted points to 0.0002; the MMRV cannot — likely computed over the 7 design-implied points the panel leaves undrawn, which no reader can recover (§7.2)

The last row is deliberate. This paper applies its own checks to itself, and they fired: a granularity verdict withdrawn after computing its false-pass rate, survey counts corrected in both directions on source verification (§8.0), a preregistered pipeline that silently dropped two of its own requirements (§9). We publish the corrections rather than the clean version because the failure mode we describe, a stated procedure that quietly does not match what was computed, does not spare the people writing about it.

1.1 Related work

Two audit traditions come before this paper; neither asks our question.

Instrumentation audits in RL and robotics. *Deep Reinforcement Learning that Matters* (arXiv:1709.06560) showed that reported RL gains often failed to survive changes in seed, environment, or codebase. *Deep RL at the Edge of the Statistical Precipice* (arXiv:2108.13264) showed that point-estimate comparisons at small run counts are unreliable and shipped rliable to fix it. Both established the pattern we follow: an instrumentation paper earns its place by re-adjudicating named numbers and shipping a tool.

The closest work, and the gap it leaves. *What Are We Actually Benchmarking in Robot Manipulation?* (arXiv:2606.04233) audits four failure modes in manipulation benchmarks: shortcut solvability, lack of statistical significance, creeping overfitting, and data-source dependence. Using paired task-stratified Wald tests it finds that only 19.8% of LIBERO and 19.7% of SimplerEnv SOTA claims are provably significant, and that a probe with no language encoder scores at or near reported SOTA.

It also identifies, independently and in stronger language than we do, the selection problem of our §5:

“The score we report is the best of several checkpoints, each scored on the reported suite... Keeping the checkpoint that scores best on it is therefore selection on the test set.”

They apply that critique to their own probe and make the practice visible. Our §5 shows the same choice flips the sign of a published correlation on one task, and that no surveyed paper states which rule it used.

The gap it leaves is the space this paper occupies. That audit contains no mention of *correlation*, *real-to-sim*, *real2sim*, or *sim-to-real*. It asks whether a benchmark *score* supports the claim that policy A beats policy B. We ask whether a *sim-real correlation* supports the claim that the simulator can stand in for hardware at all. Two routes reaching the same place is stronger evidence than either alone.

What neither line covers. Both that audit and *reliable* analyze data the benchmarks publish. The papers we examine publish no machine-readable results at all (§3), so the prior question, *can this number be checked by anyone?*, has not been asked in this setting.

The statistics are a century old, and we claim no novelty in them. Drop-one is a case-deletion influence diagnostic (Cook 1977; Belsley, Kuh & Welsch 1980). The interval is Fisher’s small-sample z (Fisher 1915; 1921). The permutation floor is the discreteness of the randomization test (Fisher 1935; Pitman 1937), and the bootstrap ceiling is elementary counting over resamples (Efron 1979). Computing significance over pooled dependent points is pseudoreplication (Hurlbert 1984), which shrinks the effective sample by the design effect (Kish 1965); §13(b)’s attenuation logic goes back to Spearman (1904), with its modern small-sample complication in Loken and Gelman (2017). The contribution is the audit: recovering the data these classical tools need from a literature that does not release it, and reporting what they show.

2. What to report

Five checks. Each is arithmetic on data the authors already have, and each is included because it catches something the conventional reporting in this literature misses.

check	question it answers	when it fires
Leverage	Is one unit carrying the correlation?	max drop-one $ \Delta r > 0.10$
Fisher-z	What can this many independent units support?	always — report the interval
Exact	Is a test at $\alpha = 0.05$ even possible?	$k \leq 3$, where $\min p = 1/k! > 0.05$
permutation		
Bootstrap	Is a percentile CI meaningful?	$C(2k-1, k) < 50$ attainable values
support		
Granularity	Is your own reported value attainable at your stated design?	value is not a multiple of $1/(N \cdot n)$

The first matters most, and nothing in current practice provides it.

(The $1/k!$ minimum is for a one-sided test. A two-sided test can require up to $2/k!$ (0.083 at $k = 4$), but only when some permutation attains the observed $|r|$ with the opposite sign. On the *Digital Cousins* data none comes close: all 24 block permutations give positive r ; the largest competitor is 0.871, and the attained two-sided p is $1/24 = 0.0417$. §8’s $k \leq 3$ verdicts only strengthen under a two-sided reading.)

2.1 The same tool, two published papers, opposite verdicts

The tool’s output on both, verbatim:

RoboWorld, Fig. 9a (GPT-4o score)

8 points over 8 independent units

$r = +0.9888$ reported 0.989

leverage max drop-one $|\Delta r| = 0.191$ [0.798, 0.994] ONE UNIT CARRIES THIS RESULT

Fisher-z 95% [+0.937, +0.998] (df = $k-3 = 5$)

exact perm. min attainable $p = 1/8! = 0.0000$ test at .05 possible

bootstrap $\leq C(15,8) = 6435$ distinct values usable

→ Report the drop-one range 0.798–0.994 alongside r ; one unit moves it by 0.191. Check the rank correlation too – leverage on Pearson need not affect ranking.

Digital Cousins (unit = architecture)

16 points over 4 independent units

$r = +0.9094$ reported 0.91

leverage max drop-one $|\Delta r| = 0.013$ [0.905, 0.922] no single unit dominates

Fisher-z 95% [-0.410, +0.998] (df = $k-3 = 1$)

exact perm. min attainable $p = 1/4! = 0.0417$ test at .05 possible

bootstrap $\leq C(7,4) = 35$ distinct values too coarse for a percentile CI

→ Do not report a bootstrap CI: only 35 distinct values are attainable at $k=4$.

RoboWorld has the larger sample and a p -value. Digital Cousins has neither, and half the independent units. The conventional signals rank them in exactly the wrong order.

Two things are worth reading off these outputs. A high r with a small p can still rest on one point: RoboWorld’s $p < 0.001$ is correct and says nothing about the drop-one range. And Digital Cousins’ r is stable to unit composition, yet its k -unit interval is wide: Fisher-z (the textbook way to put an interval on a correlation) spans [-0.410, +0.998] because $k = 4$ genuinely supports little.

2.2 The choice the checks depend on

Every check needs to know what an independent unit is, and that is where the literature is silent.

Successive checkpoints of one training run are not independent samples. Neither are task-policy cells sharing a policy. No paper in our survey states which it treats as its unit, and the choice moves the numbers substantially: on Digital Cousins, dropping a *generalization level* moves r four times more than dropping an *architecture* (§4.1).

Our rule, applied throughout: **the training run is the unit; a checkpoint is not**. A reader who prefers a different rule can recompute from the released data, which is more than the surveyed papers permit.

Choosing a unit when you have tasks and policies and seeds. Evaluation data is usually crossed, not nested, so there is no single right answer. But there is a rule that decides it:

Resample the axis your claim generalizes over.

If the claim is “*this simulator predicts how a new policy will do*”, the unit is the policy, and a policy’s task cells travel together. If the claim is “*this simulator is accurate across tasks*”, the unit is the task. If you are claiming both, report both: on Digital Cousins the two give 1.9% and 8.1% (§4.1), and a reader deserves to see the larger one.

Seeds, where they exist, are the cleanest unit available and should be preferred over either. Almost no paper in this literature has them.

2.3 Cost

`correlation_audit.py`, ~280 lines, numpy and scipy only. Runs in milliseconds on any of these datasets. The inputs are the scatter points and one unit label per point, quantities every author already has and none currently reports.

3. Recovering the data

No paper we surveyed publishes machine-readable evaluation results. (Throughout, the **subject paper** is `real2sim-eval`, arXiv:2511.04665, the paper whose data we recover and re-analyze most deeply.) We checked paper text, project sites, and repositories: `SimplerEnv` (1,126★) and `PolaRiS` (221★, both as of 2026-07-20) ship simulation environments

and code but no result files; REALM’s paper promises “*full non-aggregated results... on the project website*”, which links a repository containing 59 YAML files (58 configuration, one CI workflow) and no results. Across every source tree there is not one CSV or JSON of results.

So we recover coordinates from the published figures. Many are vector artwork, so the plotted values are exact drawing commands rather than pixel estimates, and the arXiv e-print tarballs often contain original vector figures even where the compiled PDF rasterizes them.

Recoverability: 8 of 12 papers attempted. The remainder failed for reasons no effort fixes: three have raster figures (verified in their e-print sources: two ship PNGs, one wraps a raster inside a PDF container), one has no scatter plot. Of the ten papers added to the survey after this pass (six found during earlier drafting, four via the completeness searches of §8.1), three (Cosmos-Surg-dVRK, DreamDojo, MolmoSpaces) ship vector scatter figures that we extracted and validated against their printed statistics (§4.2 reports their leverage results), bringing the recovered set to eleven of twenty-two. REALM, one of the original eight, was also re-extracted point-by-point across all four of its panels (data/realml.csv). EmbodiedSplat and Mem-World ship raster figures only. *Our first five attempts were hand-picked as likely to succeed and returned 4/5; the unselected remainder returned 3/6; the subject paper’s own figure, extracted before either pass, is the twelfth attempt and eighth success. We report the pooled 67% (8/12), not the selected 80%.* Pooled over all seventeen attempts to date (the original twelve plus five papers attempted as the survey grew), recoverability is 11/17 (65%).

Validation. Every dataset used here reproduces a statistic its authors published (in seven cases an r printed inside the figure itself) to ≤ 0.0006 (Digital Cousins’ figure prints the rounded 0.91; our 0.9094 additionally matches that paper’s numeric tables exactly). Several also land exactly on the k/n lattice implied by the stated trial count (a success rate over n trials can only be a multiple of $1/n$).

3.1 What an extraction can and cannot be validated against

We validated every recovered dataset by reproducing a statistic its authors published, and that licenses less than it appears to. Pearson r averages over point-identity errors; a maximum-type statistic does not. Two cases taught us this: one where a published figure’s plotted positions deviate from the same paper’s own numeric table, and one where our own r -validated data failed to reproduce the same paper’s MMRV.

An extraction must be validated against a statistic with the same conditioning as the one you intend to compute. An r -based validation licenses r -based claims and nothing else.

Every r -based finding in this paper is covered by that validation. Appendix A records both cases in full, including the resolution of the MMRV mismatch.

(A validation stronger than reproducing one statistic is occasionally available: on the subject paper’s Figure 9 our extracted whiskers match exact 95% Clopper–Pearson intervals to $<10^{-5}$ on all 208 bounds, which recovers the episode counts themselves from the figure geometry alone: $\text{sim } n^ = 200$, real $n = 20/27/16$ per task.)**

We also bounded the extraction error itself with a fully independent method: a pixel-level re-extraction of RoboWorld Fig. 9a (300 dpi render; legend-sampled color-blob detection, gridline-based calibration; pixel_reextract.py, released) reproduced all eight points to within 0.61 x-units and 0.0013 y-units (the worst case an occluded marker), changing r by 0.0001 and the drop-one range not at all. Occlusion, not calibration or anti-aliasing, is the dominant pixel-method error source; the vector method resolves occluded and exactly coincident markers that pixel methods cannot.

4. Why the leverage check: one policy can carry a correlation

This is the evidence for §2’s **leverage** check, which asks how much one data point moves the answer. Removing one independent unit at a time from RoboWorld, one of the survey’s largest designs ($k = 8$ policies, reported $p < 0.001$):

panel	published r	drop-one range	range width	$\max \Delta r $ from r	most influential unit
9a GPT-4o score	0.989	[0.798, 0.994]	0.196	0.191	PaliGemma Binning
9b Gemini score	0.944	[0.778, 0.969]	0.190	0.165	PaliGemma Binning
10a GPT-4o success rate	0.901	[0.836, 0.975]	0.139	0.075	Pi0.5 — <i>not</i> the isolated point
10b Gemini success rate	0.876	[0.708, 0.971]	0.263	0.169	PaliGemma Binning

The range width and the max $|\Delta r|$ are two different statistics, and an earlier draft of this table conflated them. The *range width* is the gap between two different removals; the max $|\Delta r|$ is how far a single removal moves the published value, and it is the quantity §2.1 reports and the tool computes. They nearly coincide on 9a (0.196 vs 0.191), which is what hid the difference. On 10a they do not: no policy moves r by 0.139; the largest single effect is 0.075.

On two of four panels, one policy carries the correlation. Removing PaliGemma Binning takes 9a’s r from 0.989 to 0.798, while the other seven removals move it by at most 0.005, a 38× ratio. The same holds on 9b (0.165 vs 0.025). That policy sits at $x = 797$ against a cluster spanning 1474–1621.

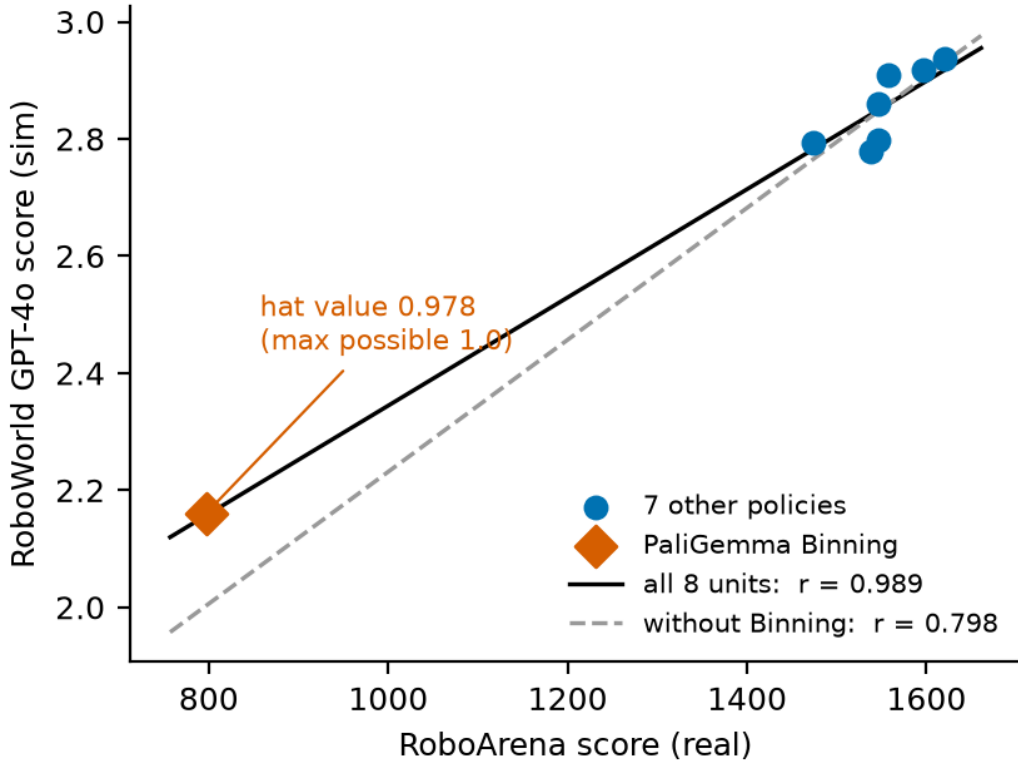


Figure 1. RoboWorld Fig. 9a, re-plotted from our extracted data. Removing PaliGemma Binning (diamond, hat value 0.978 of a maximum 1.0) moves r from 0.989 to 0.798; the other seven removals move it by at most 0.005. Nothing conventionally reported, neither $p < 0.001$ nor the Fisher-z interval, distinguishes the solid fit from the dashed one. Regenerated by `figures/make_figures.py` from `data/roboworld.csv`.

It does not hold on the other two panels, and the exception is instructive. On 10a the most influential policy is Pi0.5 (0.075), not the isolated point (0.065), so the geometric story in §4.1 is a tendency, not a law. On 10b Binning

leads but only by 1.8×. We report all four rather than the two that make the cleanest case.

The durable fact underneath is a property of the design. PaliGemma Binning has a hat value of 0.978 out of a maximum 1.0, very near the theoretical ceiling for a single point in an eight-point design (hat values sum to 2 in simple regression, so no reading of “98% of total leverage” is intended). That holds for the figure itself, independent of any statistic computed from it, and it is what makes three decimal places on $r = 0.989$ optimistic.

The check that complicates this, which we ran

RoboWorld’s rank correlation is stable to the same removal. Spearman $\rho = 0.9701$ on all eight policies; dropping PaliGemma Binning gives $\rho = 0.9550$, a change of 0.015, and dropping any of three other policies changes it by the same amount.

A corollary we adopt as a recommendation: for score-valued axes, the rank correlation is the defensible headline statistic, with Pearson beside it if desired.

The accurate statement is therefore narrower than “the result is fragile”: RoboWorld’s Pearson estimate is leverage-dominated, and its ranking claim, the claim a simulator is really sold on (§7), is not.

It also sharpens what the leverage check is for. It fires on Pearson, an interval-scale statistic, computed here over a leaderboard score whose interval scale is not obviously meaningful. **A leverage warning is a prompt to check the rank correlation, not a verdict.**

A reported p -value does not surface this: $p < 0.001$ certifies the correlation is non-zero, not that it survives removing one of eight policies — and here it does not. Nor would a point-resampling bootstrap: resampling points reproduces the leverage rather than diagnosing it. Nor does the Fisher-z interval, whose width is a function of r and k alone and cannot see point geometry: 9a’s interval is [0.937, 0.998], and nothing in it warns that removing one of the eight units lands at 0.798, far below its lower endpoint. The checks are complementary. When the leverage check fires, read the Fisher-z interval as what k well-behaved units would support, not as coverage for this design.

4.1 ...and the check correctly clears Digital Cousins

clustering axis	k	drop-one swing	as % of r
policy (architecture)	4	0.017	1.9%
generalization level	4	0.074	8.1%

$r = 0.9094$ (published 0.91), and no architecture moves it by more than 0.013. **Composition sensitivity is not a universal property of small- k correlations.** It depends on point *geometry*: RoboWorld has an isolated point at the end of a flat range; Digital Cousins spans 10–70% evenly.

So the diagnostic is not “*is k small?*” but “**is the correlation carried by an isolated point?**” That is a question any author can answer in one line, and none of the surveyed papers asks it.

4.2 The check across the twenty-two released datasets

Two examples is an anecdote. Here is every unit-labeled dataset in our released CSVs (Digital Cousins under both defensible unit choices; the last six extracted from papers added later to the survey), with the drop-one leverage and the same statistic on Spearman (since §4 shows Pearson leverage need not affect ranking). Drop-one is over independent units where they exist; rows marked *pts* have no independent units (or too few) and drop single points instead:

dataset	k	r	max $ \Delta r $	fires?	max $ \Delta \rho $
real2sim-eval toy	4	0.9444	0.006	no	0.054
real2sim-eval rope	4	0.9007	0.029	no	0.073
real2sim-eval T-block	4	0.9147	0.048	no	0.090

dataset	k	r	max $ \Delta r $	fires?	max $ \Delta p $
RoboWorld 10a	8	0.9005	0.075	no	0.042
RoboWorld 10b	8	0.8762	0.169	yes	0.113
RoboWorld 9a	8	0.9888	0.191	yes	0.021
RoboWorld 9b	8	0.9438	0.165	yes	0.090
Digital Cousins (unit = policy)	4	0.9094	0.013	no	0.047
Digital Cousins (unit = gen. level)	4	0.9094	0.037	no	0.063
Cosmos-Surg-dVRK automated	3	0.7560	0.073	no	0.086
Cosmos-Surg-dVRK manual	3	0.7180	0.124	yes	0.131
DreamDojo	6 pts	0.9953	0.010	no	0.057
MolmoSpaces pick	8 pts	0.9585	0.020	no	0.000
MolmoSpaces open	4 pts	0.8523	0.617	yes	0.900
MolmoSpaces close	4 pts	0.9727	0.040	no	0.000
REALM Overall	3	0.9187	0.025	no	0.022
REALM Default	3	0.8838	0.135	yes	0.057
REALM VB-POSE	3	0.9335	0.080	no	0.187
REALM V-VIEW	3	0.8902	0.168	yes	0.080
subject paper toy, 200 episodes	4	0.8970	0.011	no	0.095
subject paper rope, 200 episodes	4	0.9181	0.043	no	0.088
subject paper T-block, 200 episodes	4	0.9501	0.037	no	0.071

Median max $|\Delta r| = 0.045$; range 0.006–0.617.

The check fires on 7 of 22, across four papers and four different evaluator designs. An earlier draft could report firings only within RoboWorld, and a reader raised the strongest objection to the check in that form: it might be separating “this design contains an isolated point” from “it does not,” rather than fragile from robust, since all three firings shared one geometry. The extension answers this. Cosmos-Surg-dVRK’s manual panel fires with no isolated point at all: dropping the $\pi 0$ run (eight of its twenty-four points, which travel together under §8.1’s Rule 1) moves r from 0.718 to 0.594. MolmoSpaces’ open panel fires at $k = 4$ points with the largest swing in the set (0.617 on Pearson, 0.900 on Spearman), an appendix panel that inherits the same checks. And two of REALM’s four panels fire on a third mechanism: GR00T N1.5’s drawer-task scores sit at or near zero on both axes, so that one policy anchors each correlation: dropping it moves r by up to 0.168 in panels that print $p < 0.001$. The subject paper’s 200-episode rerun, at the bottom of the table, passes everywhere at both episode scales (§13(b)).

Calibration against a healthy null. A firing alone does not establish an aberrant unit, because at small k even healthy data moves. We simulated designs with no aberrant unit at all, matched to each firing panel’s printed r , unit count, and points per unit (leverage_null.py, seed-fixed, released). RoboWorld’s 0.191 swing at $k = 8$ sits at the 99.8th percentile of its healthy null ($P = 0.002$): a genuine anomaly, not smallness. The extension firings read differently: Cosmos-Surg-dVRK’s 0.124 at $k = 3$ ($P = 0.69$), REALM’s swings at $k = 3$ ($P \approx 0.54$ – 0.57), and MolmoSpaces’ 0.617 at $k = 4$ ($P = 0.28$) all sit inside the range that healthy designs of their size produce. The mechanisms named above are real as descriptions of where each swing comes from; their statistical surprise is low. Read one way this tempers the extension. Read the other it is the paper’s thesis again: at three units, stability cannot be demonstrated even when present, and the 0.10 flag partly measures k itself — with a true correlation of 0.9 and one point per unit, healthy designs exceed it 52% of the time at $k = 4$, 29% at $k = 8$, and 4% at $k = 16$. The drop-one range remains the report; the percentile against a matched null, now computable for any design, is its calibrated reading.

Three things follow.

1. **A dominant unit is the failure mode.** RoboWorld’s isolated point at the leverage ceiling (hat value 0.978 of a maximum 1.0; null percentile 99.8) is the clean case: one line identifies a place where published decimals mislead. The run-level and zero-anchor firings (Cosmos, REALM) show the same arithmetic catching a different design property, with the calibration above saying how much of each swing is the design and how much is k .
2. **It is cheap and it clears most datasets.** Fifteen of twenty-two pass. A diagnostic that fires on a third and points at a real design property is a filter, not an alarm. The 0.10 threshold is not tuned: any cutoff between 0.09 and

0.12 selects the same seven firings. And 0.10 is roughly the smallest correlation difference the surveyed papers themselves interpret: the subject paper’s fidelity ablations move r by 0.14–0.39, and A Practical Recipe reports a $0.875 \rightarrow 0.970$ sensitivity gain as an improvement.

3. **The rank column shows it is not redundant with ranking stability.** RoboWorld 9a has among the largest Pearson leverages in the set (0.191) and the *smallest* rank leverage (0.021); MolmoSpaces open moves both; REALM VB-POSE is the converse: Pearson passes (0.080) while Spearman moves 0.187. The two answer different questions, and reporting only one hides the other.

We do not claim the check generalizes beyond identifying the design property it measures; the extension shows that property is not confined to one paper or one geometry.

5. Why the unit must be stated: an unstated choice flips a sign

Which checkpoints enter the correlation is a choice, and it can flip the sign of the answer. Papers vary in whether they correlate *all* evaluated checkpoints or one representative per policy; the choice is rarely stated and never justified:

task	all checkpoints	best checkpoint per policy (n=4)
toy packing	0.9444	0.9410
rope routing	0.9007	−0.3143
T-block	0.9147	0.8778

A published r of +0.90 becomes −0.31 under a defensible alternative rule. Both columns come from the same recovered dataset, so this is a selection effect, not a provenance one. An independent human transcription of the same figure reproduces the sign reversal (−0.19).

Our own selection rule, stated: “best checkpoint” here means highest real-world success rate, first-listed row on ties, itself one of several defensible rules. Ties alone span rope from −0.47 to −0.10; selecting by highest *simulated* success (arguably what a practitioner deploying from sim would do) gives −0.68. Every variant preserves the sign reversal; the value printed depends on the rule, which is the section’s point applied to ourselves.

The effect is rope-specific; the other two tasks barely move. We therefore do *not* claim “correlation at $n=4$ is uninterpretable,” which would over-generalize from one of three tasks.

We decline to settle which estimand (which quantity the correlation is actually claiming to measure) is the right one: the all-checkpoint r is pseudo-replicated (≈ 20 points from 4 runs, dependent points counted as if independent), but the best-checkpoint r has $n = 4$ and corresponds to what a practitioner actually does, and the literature has not addressed the choice. The practical recommendation follows from §2.2’s rule regardless. The two choices are different claims: all-checkpoints speaks to within-run tracking (when to stop training), best-per-run to across-run selection (which policy to deploy). **State which claim is being made; when both are, report both correlations:** here +0.90 and −0.31. For uncertainty, the unit stays the training run in either case: checkpoints are serially dependent draws whichever estimand they enter.

6. Why Fisher-z and permutation, not bootstrap: uncertainty at small k

At these sample sizes, resampling or shuffling can produce only a handful of distinct outcomes: the counting argument behind §2’s **bootstrap-support** and **permutation** checks. Resampling k independent units with replacement gives k^k draws, and Pearson r depends only on the multiset drawn, so the bootstrap distribution has at most $C(2k-1, k)$ distinct values. This requires no data recovery; it is arithmetic on a count the papers state.

k	papers	ceiling	mass of one draw
1	RoboSnap, SC3-Eval, DreamDojo, dWorldEval	1	100%
2	WEAVER, EmbodiedSplat, Mem-World	3	25%
3	REALM, WorldGym, Cosmos-Surg-dVRK	10	3.70%
4	real2sim-eval, PolaRiS, SIMPLER, Digital Cousins, WorldEval, MolmoSpaces	35	0.39%
5	A Practical Recipe, SimFoundry [†]	126	0.032%
8	RoboWorld, Gemini/Veo	6,435	0.000006%
18	PlayWorld	4.5×10^9	$\sim 10^{-21}$ %

[†] SimFoundry’s headline is a mean of seven per-task correlations at $k = 3$ (finetuned tasks) or $k = 5$ (zero-shot); we code it at 5, the conservative direction.

The ceiling is attained only for data in general position: with singleton units it falls to 15 at $k=4$, because a resample like $\{a,a,b,b\}$ has two distinct locations and r on two points is ± 1 regardless of values. We report ceilings, which is the conservative direction.

This is a small-sample result and it dies at large k . Strong for the eleven papers at $k \leq 5$; no result at all for RoboWorld at $k = 8$.

6.1 What still works at $k = 4$, and what to use instead

Uncertainty is quantifiable at these sample sizes, just not by the method most likely to be reached for. Discreteness defeats the cluster bootstrap and the jackknife; Fisher-z, exact permutation, and a Bayesian posterior all remain valid. Verified at $k = 4$ on our data:

method	status
Cluster bootstrap	≤ 35 atoms; lower endpoint set by 2–3 resamples of 256
BCa	unstable — acceleration from 4 jackknife values, z_0 from a 35-atom distribution
Fisher-z	works — toy $[+0.431, +1.000]$, rope $[-0.800, +0.993]$
Exact permutation	works — $4! = 24$ labelings, minimum $p = 0.0417$

The recommendation is specific: at $k \leq 5$, report Fisher-z and an exact permutation p , and do not report a bootstrap CI. Fisher-z will often span most of $[-1, 1]$; that is the correct output, not a failure of the method: an interval that wide is what $k = 4$ supports, and it is more useful than a bootstrap interval that looks tighter only because it has 35 places to land.

(A percentile interval remains formally defined at any k ; what fails is coverage. On the three checkpoint tasks only 1–3 atoms, the distinct values resampling can produce, lie below the 2.5% cutoff, and on Digital Cousins four of 256 resamples do, so a publishable-looking interval rests on two to four resamples.)

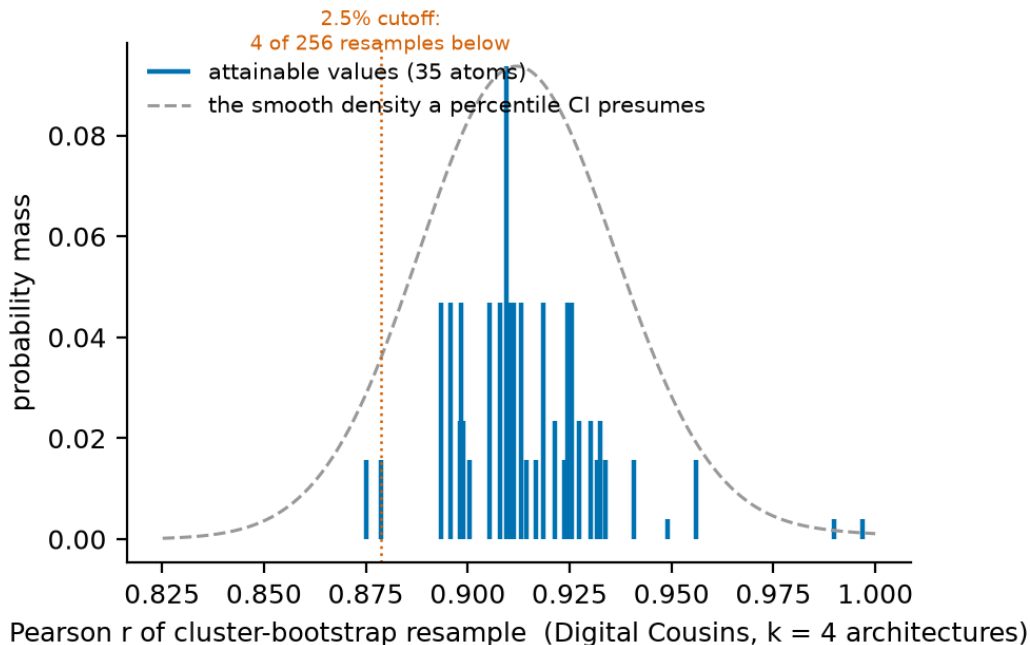


Figure 2. The entire cluster-bootstrap distribution on Digital Cousins ($k = 4$ architectures): 35 attainable values. Four of 256 resamples fall below the 2.5% cutoff, so a percentile interval’s lower endpoint rests on four draws. The dashed curve is the smooth density such an interval presumes. Regenerated by `figures/make_figures.py` from `data/digital-cousins.csv`.

A note on how we compute Fisher-z when points outnumber units. Where each unit contributes one point (RoboWorld, $k = 8$), the interval is the standard Fisher-z CI. Where it does not, there are two options and we use both deliberately: the Digital Cousins interval in §2.1 centers the pooled 16-point r and takes df from $k = 4$ units, a reference bound on what k units could support rather than a calibrated CI, since no sampling scheme yields exactly that distribution. The intervals above are instead computed after aggregating checkpoints to their per-run means (unit-level r : toy 0.984, rope 0.697; these differ from the pooled all-checkpoint values in §4.2). Aggregation is the cleaner convention; the pooled-center variant is the more cautious one on these data, and either must be stated, which no surveyed paper does.

How uncalibrated is it? Measured (fz_coverage.py, released; 4 clusters $\times$ 4 points, cluster effects on both axes, 10,000 replicates per condition): the pooled-center bound covered the unit-level correlation in 97–100% of replicates across $p \in \{0, 0.5, 0.9, 0.95\}$ — conservative rather than calibrated, median width 0.95–1.9 on the $[-1, 1]$ scale. The naive pooled Fisher-z with $n = 16$ ($df = 13$) covered it in only 59–80%, dropping to 4–20% when within-cluster noise carried no signal: the false precision pooling non-independent points produces. The bound’s one failure mode is leverage (displacing one cluster by 3 sd cut its coverage to 87.5% in the worst condition): exactly the regime where §4’s leverage check fires, and why the parenthetical pairs them.

7. Why the checks apply to MMRV too — where they matter more

Most surveyed papers report **MMRV** (Mean Maximum Rank Violation) alongside r . It is closer to the actual claim, since these simulators are sold for deciding which policy to deploy, and its stability has not been examined.

Drop-one swing as a percentage of each metric’s own value:

Swing = (max – min of drop-one values) / the full-panel value, identically for both metrics; violation pairs are strict sign disagreements, magnitudes are real-side, and RoboWorld’s rescale is fixed from the full panel.

dataset	k	MMRV	abs swing	r swing	MMRV swing	ratio
Digital Cousins, by policy	4	0.105	0.029	1.9%	27.7%	14.6×
Digital Cousins, by generalization level	4	0.105	0.097	8.1%	92.4%	11.4×
RoboWorld 10a*	8	0.0058	0.0035	15.4%	60.2%	3.9×
RoboWorld 10b*	8	0.055	0.046	30.0%	83.4%	2.8×

* RoboWorld’s x -axis is a leaderboard score, so MMRV’s $|R_i - R_j|$ is min-max rescaled, an analogue rather than their metric. We print the absolute swings beside the percentages because MMRV values near zero make percentage swings large by construction; RoboWorld 10a’s 60.2% is 0.0035 in the metric’s own units. One convention footnote: this table excludes tied pairs from violations (sign product < 0) and uses real-side gaps, SIMPLER’s definition; under a tie-inclusive reading Digital Cousins’ full-panel MMRV would be 0.110–0.112 (depending on how ties are read) rather than 0.105, and §7.2 shows the subject paper uses simulated-side gaps entirely. Recomputing this table under the subject paper’s convention changes the Digital Cousins swings to 42.8% and 52.2%: different values, same conclusion (still 22× and 6× the r swings). No convention choice affects any swing conclusion, but the conventions are not interchangeable and no paper states which it uses.

The sharpest case is Digital Cousins, precisely because the leverage check clears it (§4.1). Its correlation is stable while its MMRV moves 28% under the same perturbation. The two metrics come apart; a reader checking only r would call the result robust.

Mechanism. MMRV is a mean with many exactly-zero terms: an item contributes only if some pair involving it has a rank violation. Measured, 25–75% of items contribute exactly zero, depending on the dataset. Instability is items switching between 0 and a large real-side gap: discontinuous, not drift.

We initially argued that a max over pairs lets one point dominate. That is wrong. The max is over $|R_i - R_j|$, a real-side quantity anchored by real-side extrema, which is stabilizing; zero-inflation is what the data supports.

7.1 A granularity check — and the one verdict it supports

A published MMRV can be checked with no data at all: it can only take multiples of $1/(N \cdot n_{\text{ep}})$, so it must land on that lattice within 3-decimal rounding slack. An earlier draft claimed this test is “discriminating in every case examined.” That was wrong, and a reader caught it: the test’s power depends entirely on the lattice spacing, and we had not computed its false-pass rate.

For an arbitrary value the pass probability is $\min(1, 2 \cdot \text{slack}/g) = \min(1, 0.001 \cdot N \cdot n_{\text{ep}})$:

paper	N	n_{ep}	granularity	an arbitrary value passes	usable?
real2sim-eval (T-block)	12	16	0.00521	19.2%	yes
WorldEval	4	40	0.00625	16.0%	yes
RoboSnap	10	30	0.00333	30.0%	weak
real2sim-eval (toy)	17	20	0.00294	34.0%	weak
SimFoundry	21	25	0.00190	52.5%	no — and inapplicable
real2sim-eval (rope)	20	27	0.00185	54.0%	no
SC3-Eval	21	36	0.00132	75.6%	no

We therefore withdraw the SimFoundry verdict. Its MMRV of 0.018 misses the nearest lattice point by 0.00086 at its stated $N = 21$, but an arbitrary number passes that lattice 52.5% of the time, so a miss is weak evidence of anything. Monte Carlo over 200,000 draws confirms 52.6%. Source verification later showed the test was inapplicable outright: SimFoundry’s 0.018 is a mean of seven per-task MMRVs (29 policy-task points, $k = 3$ or 5 per task), and a mean of values from different lattices lies on no single lattice. The false-pass arithmetic above, computed for a single $N = 21$ lattice, was moot before it was weak.

One verdict survives, as support for the extraction rather than as a fault. `real2sim-eval` prints no checkpoint count for Table I itself; the only count printed anywhere is the appendix’s “16/15/12 checkpoints” sentence, which describes its replay experiment (§8.1) and is the closest a reader gets to a stated N . At that $N = 12$, the T-block MMRV of 0.108 misses the lattice by 0.00137, where only 19.2% of arbitrary values pass. At the $N = 15$ our extraction recovered from Figure 3, the miss is 0.00033, inside the slack. The arithmetic favors the recovered count over the appendix’s: independent support for §3’s extraction on the task where we most needed it, and consistent with the appendix sentence describing a different evaluation than the one behind Table I.

The general lesson for the check. Granularity is only informative when $N \cdot n_{\text{ep}} \lesssim 300$. Above that the lattice is finer than reporting precision and the test says nothing. `correlation_audit.py` now reports the false-pass rate alongside every granularity verdict, so this cannot be quoted without its own power.

7.2 The ranking metric’s convention must be brute-forced — and in one paper it cannot be

We could not at first reproduce the subject paper’s Table I MMRVs from points that reproduce its r to four decimals, and an earlier draft attributed the residual to simulator re-run noise. That explanation was wrong, and finding the right one is the finding. The values reproduce **to print precision on all three tasks** — but only under a specific undocumented convention: a rank violation whenever the $\leq$ -orderings disagree, weighted by the *simulated-side* gap, the reverse of SIMPLER’s defining equation. The same convention reproduces the paper’s 200-episode appendix figure exactly, as rational lattice points 21/200, 307/2000, 209/3000 (our `data/real2sim-eval-fig9-200ep.csv`, r validated to ≤ 0.00014). The subject paper’s metric is internally consistent and fully recoverable.

The catch is what recovery required. The released repository contains rollout and success-scoring code but no implementation of MMRV or the correlation, so the convention could be identified only by brute-forcing violation and weighting variants until one matched two figures at once. The grid is released as `mmrv_conventions.py`: 60 variants (five violation readings $\times$ four gap sides $\times$ three normalizations), of which exactly one matches Table I on all three tasks, and the same one matches the appendix figure as exact fractions. Our own first three attempts, each checking a plausible pair of conventions, wrongly concluded the values were unreproducible. **The field’s ranking metric is being computed under at least two argument orders (SIMPLER’s definition weights the real-side gap; its most transparent adopter weights the simulated side), and no paper states which, or releases the code that would settle it.**

One panel resists even brute force, most likely because part of its data is not drawn. REALM’s V-VIEW panel prints $r = 0.89$ and $\text{MMRV} = 0.253$; our extraction reproduces the r from the panel’s plotted points to 0.0002. But the panel plots only 14 of the 21 points its 7-task $\times$ 3-policy design implies (the close-drawer task is undrawn for every policy), and no MMRV computable from those 14 points reaches 0.253 under any of the 60 variants in `mmrv_conventions.py`, confirmed across two independent attempts (SIMPLER’s own convention gives 0.117; the closest variant of all, an inclusive-tie reading, prints 0.244). The simplest explanation is benign: the statistics were computed over all 21 points while the panel omits the close-drawer points (a task the paper elsewhere notes sits at or near zero success), so the printed MMRV depends on seven values no reader can recover from the figure. That hypothesis is untestable from the outside, which is itself the finding, and it is the question we put to the REALM authors. Its other three panels reproduce both statistics cleanly, which is what makes the fourth diagnosable.

Appendix B records the investigation: the wrong explanation our first draft carried, the alternatives eliminated, and the convention search that resolved it.

8. Running the checks on the whole survey

Two of §2’s five checks need only k , the number of independent units, which is recoverable from every paper’s own setup description. So they can be run on all 22 surveyed papers without recovering any data:

result	papers
Cannot reach $p = 0.05$ by any permutation test	6 / 22 — REALM, WorldGym, Cosmos-Surg-dVRK ($k=3$, $\min p = 0.167$), WEAVER, EmbodiedSplat, Mem-World ($k=2$, 0.500)
<i>(check is category-inapplicable)</i>	5 / 22 — RoboSnap, SC3-Eval, DreamDojo, dWorldEval, Colosseum V2 correlate checkpoints, per-task finetunes, or perturbation conditions of a single policy or lineage, so there are no units to permute
Cannot support a percentile bootstrap CI	17 / 22 — at $k \leq 4$ the resampling distribution has ≤ 35 attainable values
Fisher-z has $df \leq 2$	19 / 22
Clears all three comfortably	3 / 22 — RoboWorld and Gemini/Veo ($k = 8$), PlayWorld ($k = 18$)

Six papers report a correlation for which no permutation test over units could reach $p = 0.05$, because the minimum attainable value, $1/k!$, exceeds it at their sample size. That permutation floor, the best p -value you could ever get by shuffling the units, is one-sided; two-sided it is at least as large, so the verdict is conservative as stated.

Earlier drafts said five, then three, then four; verifying every coding against its source and then running the documented completeness search (§8.1) settled it at six. Cosmos-Surg-dVRK joined on re-coding: its six scatter points are two checkpoint stages of three training runs, so $k = 3$ under §8.1’s Rule 1. EmbodiedSplat and Mem-World, both at $k = 2$, entered with the search. Five papers (table above) are category-inapplicable rather than failed; faulting a paper for failing a test it could not have run, and did not claim to run, is a technicality; the single-lineage designs are the sharper finding in their own right.

The framing matters here too. Three of the six (REALM, Cosmos-Surg-dVRK, and Mem-World) do print $p < 0.001$, obtained from parametric tests over pooled points; no permutation over their own independent units could attain a value below 0.167, and for Mem-World, at $k = 2$, below 0.5. (A stricter reading of REALM’s units, since π_0 and π_0 -FAST share a lineage, gives $k = 2$, which raises its floor to 0.5.) The finding is not that anyone failed a test. It is that the field routinely computes correlations over one to four independent units.

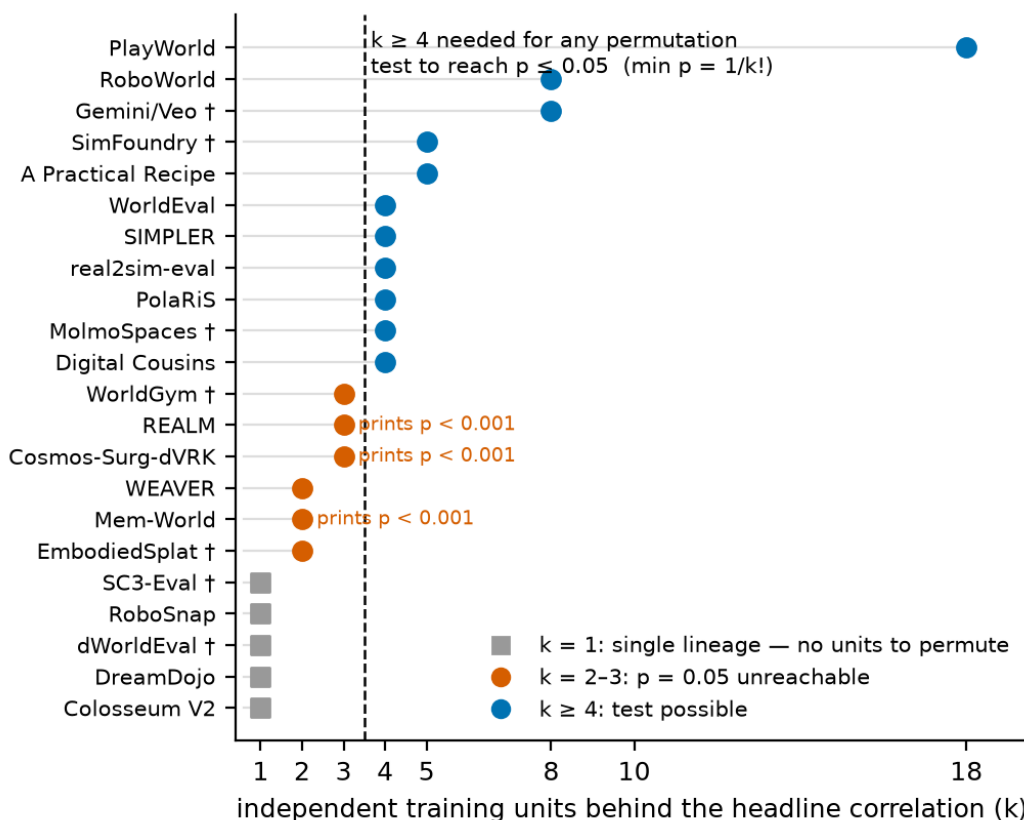


Figure 3. Independent training units behind the headline correlation, all 22 surveyed papers († = coding flagged in §8.1). Left of the dashed line no permutation test over units can reach $p = 0.05$; three of the four papers that print $p < 0.001$ sit there (the fourth, RoboWorld, is at $k = 8$). Gray squares are single-lineage designs with no units to permute. Regenerated by `figures/make_figures.py` from `survey_table.py`.

These limits are not consequences of the data being unavailable. It would be natural to read this section as downstream of §3’s finding that no surveyed paper publishes machine-readable results. It is not: $1/k!$ and $C(2k-1, k)$ are combinatorial facts about k , which follows from every paper’s own setup description. Publishing the underlying data would not add a 36th attainable value at $k = 4$, and would not let $k = 3$ reach $p = 0.05$.

The distinction matters for reading the rest of this paper:

finding	fixable by publishing the data?
We cannot reproduce one published MMRV (§7.2)	yes
We cannot check the ablation arms (§11)	yes
Our own extraction carries residual ambiguity (§3.1)	yes
6/22 cannot reach $p = 0.05$	no — this needs more policies
17/22 cannot support a bootstrap CI	no

The first three are limits on *our* verification. The last two are limits on what the *original* analyses can conclude.

8.0 The whole survey, one row per paper

A prevalence claim is only checkable if the papers are named. Every cell below was re-verified against its paper’s full text on 2026-07-20 (Colosseum V2, added by the 2026-07-21 rerun, on 2026-07-21); `survey_table.py` (released)

regenerates every count in this section from this table, so no number in §8 is hand-copied. The counts carry no confidence intervals by design: the survey is a census of a defined set, not a sample from a population, so the uncertainties that matter are coverage and coding, and both are reported (the completeness caveats and the sensitivity analysis, §8.1).

paper	k	units behind the headline r	uncertainty on r	rule stated	recovered
real2sim-eval	4/task	4 policies per task, 3–6 checkpoints each	none (per-point CIs only)	no	yes
RoboWorld	8	8 open-sourced RoboArena policies	p -values	no	yes
Digital Cousins	4	4 architectures $\times$ 4 generalization levels	none	no	yes
SIMPLER	4	6 points from 4 model lineages	none	no	yes
SimFoundry	3–5†	mean of 7 per-task r ; 29 points	none	yes	yes
WorldGym	3†	3 policies $\times$ 17 tasks	none	no	yes
RoboSnap	1	task finetunes of one $\pi 0.5$ family, 10 tasks	none	yes	yes
REALM	3†	3 VLAs ($r = 0.92$ overall, printed inside its figure)	$p < 0.001$	no	yes
PolaRiS	4	4 policies, pre-specified 1k-step checkpoints	none	yes	no — raster
SC3-Eval	1†	7 checkpoints of one architecture $\times$ 3 criteria	none	no	no — raster
WorldEval	4	4 policies	none	no	no — raster
A Practical Recipe	5	5 VLAs; correlations in tables, no scatter	none	no	no
Cosmos-Surg-dVRK	3	3 VLA runs $\times$ 2 checkpoint stages	$p < 0.001$	partial	yes
Gemini/Veo	8†	8 variants of one GROD base	none	no	no
DreamDojo	1	checkpoints of one GR00T lineage	none	no	yes
dWorldEval	1†	checkpoints of one $\pi 0$ (LIBERO headline)	none	no	no
WEAVER	2	base $\pi 0.5$ + one finetune (headline 0.870 is Spearman per its Table 8, though its text labels it Pearson)	none	no	no
PlayWorld	18	18 distinct trained policies	none	no	no
EmbodiedSplat	2†	two 4-point correlations (Polycam/DN meshes); sibling finetunes of 2 base lineages; navigation	none	yes	no
MolmoSpaces	4†	8 policy points from 3–4 lineages (CAP family, π family, Paligemma)	CIs on R and ρ, printed inside its figure	no	yes
Mem-World	2	two sibling π finetunes $\times$ 5 tasks = 10 pooled points	p-values	no	no
Colosseum V2	1	one ACT multi-task policy; points are 5–6 perturbation conditions $\times$ 3 tasks	none	no	no

† coded with a flag; the deciding ambiguity is recorded in §8.1’s cells. “Recovered” marks papers whose scatter data we extracted and validated against a published statistic: the original eight plus Cosmos-Surg-dVRK, DreamDojo, and MolmoSpaces (all in §4.2); EmbodiedSplat and Mem-World ship raster figures only. The last four rows entered via the documented completeness searches (§8.1, Scope).

Examined and excluded from every count: **GSWorld** and **Interactive World Simulator** plot a sim-vs-real scatter and print no coefficient; **AutoEval** computes and plots SIMPLER-vs-real correlations but prints no numeric coefficient for

them (its headline 0.942 is real-to-real: autonomous versus human evaluation); **RoboSimGS** reports no sim-vs-real correlation; the **benchmarking audit** (§1.1) is out of scope; **GigaWorld-1** defines the real-vs-predicted correlation as its central quantity (its Eq. 4) but never prints a value for it; its prose ps (0.78, 0.88) are video-metric-vs-evaluator-score correlations. Listed so the exclusions are auditable too.

property	prevalence
Headline r over fewer than 10 independent training units	21 / 22
No uncertainty reported on the correlation	17 / 22
Checkpoint-selection rule unstated	17 / 22

Only PlayWorld (18 policies) clears ten independent units. Three papers compute a correlation over two units. Five papers correlate checkpoints, per-task finetunes, or perturbation conditions of a single policy or lineage: one independent unit, the survey’s most extreme design fact, invisible in our earlier counts.

An earlier draft put these prevalences at 9/10, 6/10, and 9/10 over a 10-paper denominator that was never reconciled with the survey. Source verification found that every surveyed paper displays a coefficient (REALM’s numeric values, and Gemini/Veo’s headline values, appear only inside figures; Veo’s text does print one secondary OOD pair), removed two miscredited uncertainty entries (§8.2), and re-coded three unit counts; the completeness searches then grew the set from 18 to 22. The corrected prevalences are higher for sample size, lower for the selection rule: wrong in both directions, which is why this table exists.

8.1 Coding rules — several cells turn on a definition, not a fact

Rule 1. A training checkpoint is not an independent unit; the training run is. Successive checkpoints are serially correlated snapshots. This decides six papers at once (real2sim-eval, SC3-Eval, Cosmos-Surg-dVRK, Gemini/Veo, dWorldEval, DreamDojo).

Rule 2. Uncertainty must be on the *correlation*, not the points. Several papers report Clopper–Pearson intervals or standard errors per success rate and nothing on the correlation computed from them. That is one step from a bootstrap, not taken.

Rule 3. A paper reporting no coefficient is excluded. GSWorld and Interactive World Simulator each claim a correlation qualitatively (a scatter, no printed coefficient); AutoEval plots per-task SIMPLER-vs-real correlations without printing a number for any of them; RoboSimGS and one benchmarking audit report no sim-vs-real correlation at all.

Cells these rules decide:

paper	question	our coding
real2sim-eval	12–20 checkpoints, or 4 training runs per task?	4 runs per task — the recovered Figure 3 data contains all four policies on every task, each with 3–6 checkpoints (12 task-policy series; Figure 4 displays six as unrolled curves). The “16/15/12 checkpoints” sentence describes its <i>appendix replay experiment</i> ; Figure 3’s caption states no counts.
PolaRiS	Does “ <i>best-correlated checkpoint</i> ” govern its headline r ?	No. That governs the Libero-Score <i>baseline</i> . Its own procedure is pre-specified at 1k steps.
WorldGym	Is k 17 tasks or 3 policies?	Ambiguous; the paper does not say. We report 3 and flag it.

paper	question	our coding
real2sim-eval	Does Figure 9’s shaded band count as uncertainty on r ?	No. The band is the Clopper–Pearson 95% CI on each point’s <i>success rate</i> ; its caption credits it to episode count (“ <i>increasing the number of simulated episodes reduces statistical uncertainty</i> ”). That is uncertainty on the points, not on the correlation — Rule 2, applied to the subject paper.
Cosmos-Surg-dVRK	6 points, or 3 training runs?	3 runs — three VLAs each at two training stages (“one half training and one full training”): two checkpoints per run, and Rule 1 makes the run the unit.
Gemini/Veo	Are its 8 policy variants independent?	Unstated; we code 8 and flag it. All eight are variants of one GROD base; the paper never states whether they are separate training runs.
SC3-Eval, DreamDojo, dWorldEval	Any independent units at all?	One lineage each — SC3-Eval: 7 checkpoints of one $\pi 0.5$ architecture $\times$ 3 criteria; DreamDojo: checkpoints of one GR00T variant; dWorldEval: checkpoints of one $\pi 0$ for its LIBERO headline (its real-world r spans ~ 3 architectures, which we flag).
EmbodiedSplat	6 policies, or 2 lineages?	2 lineages per correlation — its four finetuned policies are sibling finetunes of the two zero-shot bases (HM3D, HSSD); each mesh variant’s 4-point correlation spans those two lineages. Its exact SRCC values (0.976, 0.866) appear only inside its Figure 1; the text prints the range 0.87–0.97.
MolmoSpaces	How many policies enter the pick correlation?	8 points, 3–4 lineages, flagged — CAP plus three CAP-EC variants (defined only in the figure legend), three π -family DROID finetunes, Paligemma Binning. Its text prints pick $\rho = 0.98$; both its included figures print 1.00, and our extraction confirms perfectly concordant ranks. The e-print ships an <i>unused draft</i> of the figure (marked “TBD”) that prints $\rho = 0.98$ — plausibly the source of the stale text value: a leftover draft, not a miscalculation.
Mem-World	Do 10 points mean $k = 10$?	No — $k = 2$. Two sibling π finetunes, trained on the same 50-episode-per-task data, evaluated on 5 tasks each; tasks are pooled as points without justification. The printed $p < 0.001$ is computed over the pooled points; the permutation floor over its two units is 0.5.
REALM	Are $\pi 0$ and $\pi 0$ -FAST separate units?	We code $k = 3$ and flag it — the two share a lineage, and the stricter $k = 2$ reading raises REALM’s permutation floor from 0.167 to 0.5 (§8).

paper	question	our coding
Colosseum V2	Do 5–6 perturbation conditions $\times$ 3 tasks give $k > 1$?	No — $k = 1$. Its hardware correlation uses one multi-task ACT policy trained from scratch ($\pi 0.5$ appears in its simulation experiments only); the points are perturbation conditions, not policies. Its Sec. IV-C lists five conditions; its Figs. 7–8 draw six. Its avg $R^2 = 0.798$ (predicting the <i>change</i> in success under a perturbation) and avg Spearman = 0.916 are averages over three tasks with no per-task values printed — so recovery was not attempted: no printed per-panel statistic exists to validate an extraction against. It states plainly that sim “ <i>results do not reliably predict the absolute success rate</i> ”; the correlation it claims is over condition shifts, and it enters under the same claim-based rule as RoboSnap.

Scope. The inclusion criterion is claim-based, not task-based: a paper enters if its validity claim is a correlation between simulated and real-world robot policy success with a printed coefficient. The set is manipulation- and world-model-dominated (21 of 22); one navigation paper (EmbodiedSplat) qualifies and is included; task type enters no check. Physics simulators, learned world models, and generated-video evaluators are pooled deliberately: the audited object is the printed correlation and the design behind it, not the evaluator’s mechanism, and every check depends on the papers’ units and points alone. The original collection was not systematic: six papers were found in two web searches. A documented completeness search on 2026-07-20 (seven logged arXiv API queries over cs.RO/cs.LG/cs.CV, 2024–2026, 30 unique hits, 20 screened candidates) added EmbodiedSplat, MolmoSpaces, and Mem-World, and confirmed the AutoEval exclusion. Two caveats: the arXiv API searches metadata, not full text, so the check has limited recall (it returned only 10 of the 22 papers known before the search: the eighteen then surveyed plus four exclusions; AutoEval, whose MMRV usage is body-text only, entered separately via a reviewer flag, not the queries); and more qualifying papers likely exist. A pre-submission rerun on 2026-07-21 (the same query battery plus four web searches) demonstrated both caveats at once: it added Colosseum V2 (posted 2026-05-26 and missed by the first search) and screened out GigaWorld-1 and RoboDojo under Rule 3 (no printed coefficient); SC3-Eval’s cell was re-verified against its v3.

Sensitivity to the coding. Rule 1 is a judgment call, so we recomputed every count in this section under two alternative codings: a permissive coding in which every evaluated checkpoint or variant is an independent unit, and a worst-case coding flipping all six contestable cells against us at once. Coding-independent: 19–21 of 22 papers report their headline correlation from fewer than ten units; at least four (REALM, WEAVER, Mem-World, and WorldGym under two of three codings) cannot reach $p = 0.05$ by any permutation; and both reporting-practice counts (17/22 no uncertainty, 17/22 no stated rule) involve no unit-counting at all. Coding-dependent: under the permissive coding the bootstrap-support count falls from 17 to 8 of 22, the number clearing all three procedures rises from 3 to 11, and the $k = 1$ category shrinks to one (Colosseum V2’s single hardware policy has no checkpoints or variants for even the permissive coding to promote), so the strongest quantitative versions of those claims rest on Rule 1, and we flag them as such. The below-floor set is nearly coding-independent: REALM and Mem-World print $p < 0.001$ below their permutation floor under every coding; Cosmos-Surg-dVRK stays below its floor only marginally under the permissive coding (minimum attainable $p = 1/6! \approx 0.0014$) and, unlike the other two, can then at least reach conventional significance.

8.2 Five papers show the reporting is feasible — and that it needs one more line

Five groups already compute and report uncertainty on a correlation at these sample sizes, which is the strongest argument that §2 is practical: RoboWorld ($p < 0.001$), REALM ($p < 0.001$), Cosmos-Surg-dVRK ($p < 0.001$), Mem-World ($p < 0.001$), and MolmoSpaces (bracketed CIs on both R and p , printed inside its figure). The remaining sixteen

papers’ silence reflects convention rather than difficulty. Three of the five, however, print p -values their own unit count cannot support under any permutation over units (§8), which is why the *kind* of uncertainty matters as much as its presence.

An earlier draft counted four. Source verification removed two miscredits, real2sim-eval’s Figure 9 band (per-point episode uncertainty, Rule 2) and a WorldGym p -value (it prints no significance statistic anywhere), and added REALM’s previously missed $p < 0.001$, giving three, before the completeness search added Mem-World and MolmoSpaces, giving five. The miscredits survived two review passes and fell only on checking the source, which is this section’s argument, made at our own expense.

What §4 adds is the missing line: RoboWorld’s $p < 0.001$ sits on a correlation that still moves far on removing one of eight policies (§4). A p -value answers “is this non-zero?”; a leverage check answers “does this survive?” The two are complementary, and the papers above are already most of the way there.

9. The same failure mode in our own work

The problem this paper describes did not spare our own pipeline: our preregistered, hash-locked analysis silently dropped two of its own requirements while reporting a clean bill of health, and the hash locks themselves did not survive intact. Both requirements are now implemented, and both produced findings the first run would have missed. We ship a preregistration linter; Appendix C gives the full account, including the linter’s own first version failing its retrospective test.

One scope statement, so the preregistration is not read as covering more than it does: it covers the original checkpoint analysis. The survey and its counts, the MMRV stability analyses, and the extraction extensions of §4.2 were not preregistered and should be read as exploratory: verified and regenerable, but not pre-specified.

10. Artifacts

Everything lives in the repository (github.com/trilamsr/research, directory `sim2real-correlation-audit/`); `README.md` documents the methodology, the equations behind every check, per-dataset provenance and validation gates, and one-command reproduction of each headline number. `correlation_audit.py` implements the five checks (`--demo` reproduces §2.1); `survey_table.py` regenerates every count in §8; `figures/make_figures.py` regenerates Figures 1–3 and prints its verification numbers. The recovered datasets ship as CSVs in `data/`, each opening with a provenance header and validated against a published statistic (§3): `real2sim-eval` (`real2sim-eval-fig3-checkpoints.csv` and the 200-episode `real2sim-eval-fig9-200ep.csv`, 52 + 52 points), RoboWorld (4 panels, 32 points), Digital Cousins (16 points), REALM (4 panels, 77 points), Cosmos-Surg-dVRK (2 panels, 48 points), DreamDojo (6 points), MolmoSpaces (4 source panels, 24 rows), and RoboSnap (10 points plus its own table, Appendix A). The preregistered analysis is `measure_noise_floor.py` with `results.json`; the preregistration itself is `PREREG-noise-floor.md` (including its hash-lock break record and v1.4 amendment), and `harness/prereg_lint.py` is §9’s linter. `fz_coverage.py` is §6.1’s Fisher-z coverage simulation (seed-fixed, 10,000 replicates per condition); `pixel_reextract.py` is §3.1’s extraction-error bound; `requirements.txt` pins `numpy` and `scipy`. `tests/` holds 26 known-answer tests plus byte-for-byte §2.1/`--demo` parity (`mmrv_conventions.py` regenerates §7.2’s convention grid; `leverage_null.py` regenerates §4.2’s null calibration; both are test-locked), and `make verify` runs the whole battery.

11. Scope

We are careful about what this does and does not establish.

- **No paper here is wrong.** Every r we examined reproduces from its own figure. The subject paper is unusually transparent: it publishes episode counts, randomization ranges, per-episode replay confusion matrices, and per-point confidence intervals. That transparency is what made this analysis possible.
- **Not that simulation-based evaluation does not work.** We measure what one statistic supports.

- **Not a claim about every paper in the field.** Twenty-two papers surveyed (six more examined and excluded), eleven recovered from figures; original collection non-systematic, extended by documented completeness searches (§8.1) whose own recall is limited.
- **No claim about the ablation deltas.** Their per-checkpoint evaluation numbers are unpublished (per-checkpoint *weights* are public on HuggingFace; the results computed from them are not); we verified this across seven locations including LaTeX sources, repositories, and 98 model and dataset repos.

Provenance. The analyses in this paper were performed with substantial machine assistance (agentic language-model pipelines) under the author’s direction. Every quantitative claim regenerates from the released deterministic scripts and data, and the verification protocol is part of the record rather than a private process: multiple independent recomputation passes, source-verified survey codings, and adversarial validation that overturned two of our own draft claims (§7.2, Appendix A).

12. What we are asking for

One line in a results table, computed from data every author already has:

$r = 0.989$ ($k = 8$ policies; drop-one range 0.798–0.994; Fisher-z [0.937, 0.998])

That single parenthetical distinguishes RoboWorld from Digital Cousins, and nothing currently reported does. (One label matters: where points outnumber units, the Fisher-z entry is a reference bound rather than a calibrated interval; §6.1 measures its coverage. Its blind spot, leverage, is exactly what the drop-one range beside it exposes. The pair is honest where neither element alone is.)

Wide intervals at $k = 4$ are not an argument against reporting them; they are the price list. At an observed $r = 0.9$, five independent runs already bound the true correlation above zero (Fisher-z [0.09, 0.99]), eight give [0.53, 0.98], and ten give [0.62, 0.98]. A field that wants a tight claim can buy one. What it cannot do is print the claim without paying.

For reviewers. Three questions to ask any sim-real correlation paper:

1. **How many independent training units (not points) is the correlation computed over, and what rule decided the unit?** Checkpoints of one run are not units. (Twenty-one of twenty-two published answers are below ten; five are one.)
2. **What happens to the correlation when the most influential unit is removed?** A p -value does not answer this; a one-line drop-one range does.
3. **Could the stated design support the inference at all?** The minimum permutation p is $1/k!$ (unreachable below $k = 4$), and a bootstrap has at most $C(2k-1, k)$ attainable values.

`correlation_audit.py` answers all three from the scatter points and one unit label per point.

And for authors at $k \leq 5$ today, the whole prescription in one box:

State k and the rule that decided the unit. Print the drop-one range. For the interval, either aggregate to unit means and report standard Fisher-z (the textbook interval, honestly wide) or report the pooled-center bound labeled as a bound (§6.1). Report an exact permutation p over units, stating sidedness. Do not print a bootstrap CI at $k \leq 5$: it has at most 35 attainable values at $k = 4$ and 126 at $k = 5$. If you report MMRV, state the violation and weighting convention or release the ten lines that compute it (§7.2).

Nor is the ask specific to physics simulators. Nine of the twenty-two surveyed evaluators are world models (WorldGym, WorldEval, dWorldEval, DreamDojo, WEAVER, Gemini/Veo, PlayWorld, Cosmos-Surg-dVRK, and Mem-World; a tenth, SC3-Eval, evaluates via generated video), and their validity case is exactly this correlation at exactly these unit counts. Nothing in the ceilings is robotic, either. Any claim of the form “*our cheap evaluator correlates with the expensive ground truth*” (an LLM judge against human raters, a learned reward model against preference labels) computed over a handful of independent units inherits the same arithmetic. $1/k!$ does not care what the units are. (We have not audited a case outside robotics; the transfer of the combinatorial ceilings is arithmetic, the transfer of the empirical findings is a conjecture.)

13. Open questions

(a) Which estimand, in theory? §5’s practical answer (state the claim; report both when both are made) leaves the question standing: all-checkpoint r is pseudo-replicated, best-checkpoint r has $n = 4$ and matches deployment practice, rope’s sign depends on the choice, and the literature has no account of which estimand a sim-real correlation should target.

(b) How should sim-side noise be modeled? Both axes are estimated from small grids, yet uncertainty is reported as if the simulated axis were exact. The obvious fix is invalid: redrawing episodes as $\text{Bin}(n, \hat{p})$ treats the observed rate as truth and attenuates r by up to 0.131 when both axes are redrawn. Classical disattenuation gives reliabilities of 0.81–0.95, and applying the correction $r/\sqrt{(\text{rel_x} - \text{rel_y})}$ returns 1.13 (toy) and 1.09 (T-block): correlations above 1 — not a measurement of attenuation but a falsification of the model that produced it, since independent binomial noise on the two axes cannot coexist with these observed r . Either the sim and real episode outcomes are positively correlated (plausible under replay evaluation, where both axes score the same trajectories), or the plug-in reliabilities are too noisy at $N = 15$ –17 to support the correction. Only rope yields a coherent corrected value ($0.90 \rightarrow 0.95$). So we do not claim the true correlations are higher than published. The design point also has a direct empirical test: the subject paper’s appendix rerun of the same three tasks at 200 simulated episodes per point, ten times the main evaluation, leaves drop-one leverage unchanged ($\max |\Delta r|$ 0.006 $\rightarrow$ 0.011, 0.029 $\rightarrow$ 0.043, 0.048 $\rightarrow$ 0.037; computed from our extraction of both figures). More episodes did not buy stability; only more units would: episode noise is at most a second-order problem next to the number of independent units, and the correction pinned at its ceiling says the observed correlations are already as high as the error budget permits.

Status

This is a pre-submission draft circulated to the authors of the audited papers for verification. Findings that only a source paper’s authors can settle (the MMRV convention behind the subject paper’s Table I, §7.2; REALM’s V-VIEW panel, §7.2; the RoboSnap figure/table difference, Appendix A) are framed as questions to them, and any response will be incorporated with attribution before submission. The completeness search (§8.1) was last rerun on 2026-07-21.

Appendices

The investigation record. Every conclusion drawn from these is stated in the main text; what follows is how it was reached and what was eliminated.

Appendix A — Extraction validation: two failures

Two cases taught us the limit of this check, and both belong in the methods rather than a footnote.

RoboSnap is the only surveyed paper printing its full numeric table inline, so it is the one case where extraction could be checked against ground truth from the same paper. It failed: extracted $r = 0.9089$ against a published 0.887 (data/robosnap.csv; both values reproduce exactly). The cause is that the figure’s plotted positions deviate from the paper’s own table: the three tasks the table ties at $\text{real} = 0.367$ are drawn coincident but at 0.340, and two further points sit 1.4–1.9 percentage points off their table values. An earlier draft attributed the gap to transposed axis labels and jitter; point-by-point extraction refutes both (the label-consistent assignment fits $56\times$ better, and transposition cannot change a Pearson r in any case). Extraction alone would have produced a plausible wrong number, because the figure itself is a slightly wrong rendering of the paper’s own data. The likely cause is mundane: a figure generated from an earlier revision of the numbers, or a deliberate nudge to keep the three tied points from overplotting. Which of these it was is a question only the RoboSnap authors can answer, and one we put to them.

Our own data. We validated by reproducing the published r to ≤ 0.0004 , then computed **MMRV** from the same points and it did not match (later resolved as a convention mismatch, §7.2), which proves the point a second way: reproducing

r cannot certify a metric whose definition had to be guessed. Pearson r averages over point-identity errors; MMRV is a maximum and does not.

§7.2 shows the MMRV mismatch has a different cause than we assumed, but §3.1’s principle stands regardless.

Appendix B — The MMRV mismatch: the wrong explanation, then the right one

The investigation record for §7.2, kept because our draft carried the wrong explanation before we found the right one.

Round one (wrong). Computing MMRV under SIMPLER’s definition (real-side gap) from points that reproduce the published r to four decimals gave 0.065/0.169/0.138 against the published 0.076/0.174/0.108. We attributed the residual to simulator re-run noise: perturbing each simulated rate by ± 1 episode and recomputing 2,000 times put every published value within ~ 1.4 sd of ours, and an earlier draft of this appendix printed a residual-over-noise table and concluded a published MMRV “cannot be reproduced from its own figure.” Two of that table’s own numbers then failed independent reverification: its r ratios (printed 0.90/0.67/0.75; nothing above ~ 0.1 is achievable under any perturbation variant tried by two independent recomputations, the true agreement is ~ 0.03 sd) and its perturbation wording (the printed MMRV ratios required a $\{-1, 0, +1\}$ jitter, not the stated ± 1), a wording-versus-code mismatch of exactly the kind §9 describes.

Round two (right). Extracting the paper’s 200-episode appendix figure supplied a second target, and a brute-force sweep over violation and weighting conventions found one that reproduces both at once: violation when the $\leq$ -orderings disagree, weighted by the simulated-side gap. Under it, Table I reproduces to print precision (0.0765/0.1741/0.1083) and the appendix figure exactly, as the rational fractions 21/200, 307/2000, 209/3000. “Consistent with re-run noise” was never proof of re-run noise; the residual was a definitional mismatch all along, invisible because three earlier searches each checked a plausible-but-wrong pair of conventions.

(Alternatives eliminated along the way: per-policy aggregation, single- and pair-point add/removes, magnitude-definition sweeps under the real-side gap. What settles it is one convention fitting two independent figures simultaneously.)

Appendix C — Our preregistration failure, and the linter

The failure mode this paper describes, a stated procedure that quietly does not match what was computed, is not confined to other people’s work. **We preregistered and hash-locked this analysis, and our first implementation silently dropped two of its own requirements** while reporting a clean bill of health. (The hash locks themselves also failed to survive intact: one was broken by the commit that shipped the linter, and one matches no blob in repository history; PREREG-noise-floor.md documents both breaks and their forensics.) The dropped requirements: a mandated comparison arm (the script was built around columns the comparison dataset lacks) and mandated robustness rows (the script hardcoded one value despite the preregistration saying “*never silently pick one*”).

Both are now implemented, and **both produced findings the first run would have missed**: the comparison arm confirms the extraction question does not affect §4’s conclusion, and the robustness rows show **29 distinct checkpoint subsets clear the reproduction gate**, making that gate partially circular for one task.

The root cause is mechanical: preregistration and code were written in sequence and never diffed. We ship a **preregistration linter** (§9) that requires each stated requirement to be claimed in code and to appear in the results file. Its first version failed its own retrospective test (run against the script that actually dropped two requirements, it reported “ok”) and was rewritten around an explicit requirement register.

References

All 30 identifiers below were resolved against arXiv and their titles confirmed to match (28 on 2026-07-20; Colosseum V2 and GigaWorld-1 on 2026-07-21).

Subject paper

- **real2sim-eval** — *Real-to-Sim Robot Policy Evaluation with Gaussian Splatting Simulation of Soft-Body Interactions*. arXiv:**2511.04665**. Every value audited in this paper is **v2's** (the current version): v2 revises both Table I's MMRV column and its appendix figure's per-panel MMRVs relative to v1 (verified by diffing the two PDFs), and it is v2 that §7.2's convention recovery reproduces to print precision. Code: github.com/kywind/real2sim-eval (MIT). Assets: [HuggingFace shashuo0104](https://huggingface.co/shashuo0104).

Papers whose data we recovered and analyzed

- **RoboWorld** — *RoboWorld: Fast and Reliable Neural Simulators for Generalist Robot Policy Evaluation*. arXiv:**2607.01060** (figures extracted from v3; v4, current as of 2026-07-21, is content-identical in the HTML e-print — figure images pixel-identical, every printed correlation unchanged). §2.1, §4, §7.
- **Digital Cousins** — *From Seeing to Simulating: Generative High-Fidelity Simulation with Digital Cousins for Generalizable Robot Learning and Evaluation*. arXiv:**2604.15805**. §2.1, §4.1, §7.
- **SIMPLER** — *Evaluating Real-World Robot Manipulation Policies in Simulation*. arXiv:**2405.05941**. Source of the MMRV metric (Eq. 1). Code: github.com/simpler-env/SimplerEnv.
- **SimFoundry** — *SimFoundry: Modular and Automated Scene Generation for Policy Learning and Evaluation*. arXiv:**2606.28276**. §7.1.
- **WorldGym** — *WorldGym: World Model as An Environment for Policy Evaluation*. arXiv:**2506.00613**. §6, §8.
- **RoboSnap** — *RoboSnap: One-Shot Real-to-Sim Scene Generation for Generalizable Robot Learning and Evaluation*. arXiv:**2607.06699**. §3.1, Appendix A (the figure-vs-own-table case).
- **REALM** — *REALM: A Real-to-Sim Validated Benchmark for Generalization in Robotic Manipulation*. arXiv:**2512.19562**. §6, §8.

Surveyed, not recovered

- **PolaRiS** — *PolaRiS: Scalable Real-to-Sim Evaluations for Generalist Robot Policies*. arXiv:**2512.16881**. Figures are raster; not recoverable.
- **SC3-Eval** — *SC3-Eval: Evaluating Robot Foundation Models via Self-Consistent Video Generation*. arXiv:**2606.18610**. Raster.
- **WorldEval** — *WorldEval: World Model as Real-World Robot Policies Evaluator*. arXiv:**2505.19017**. Raster.
- **A Practical Recipe** — *A Practical Recipe Towards Improving Sim-and-Real Correlation for VLA Evaluation*. arXiv:**2606.10366**. No sim-vs-real scatter figure exists.
- **Cosmos-Surg-dVRK** — *Cosmos-Surg-dVRK: World Foundation Model-based Automated Online Evaluation of Surgical Robot Policy Learning*. arXiv:**2510.16240**. Reports $p < 0.001$ (§8.2).
- **Gemini Robotics in a Veo world simulator** — *Evaluating Gemini Robotics Policies in a Veo World Simulator*. arXiv:**2512.10675**.
- **DreamDojo** — *DreamDojo: A Generalist Robot World Model from Large-Scale Human Videos*. arXiv:**2602.06949**.
- **dWorldEval** — *dWorldEval: Scalable Robotic Policy Evaluation via Discrete Diffusion World Model*. arXiv:**2604.22152**.
- **WEAVER** — *WEAVER, Better, Faster, Longer: An Effective World Model for Robotic Manipulation*. arXiv:**2606.13672**. Correlation over two policies.
- **PlayWorld** — *PlayWorld: Learning Robot World Models from Autonomous Play*. arXiv:**2603.09030**. The only surveyed paper with ≥ 10 independent units.

Added by the completeness search (2026-07-20; §8.1, Scope)

- **EmbodiedSplat** — *EmbodiedSplat: Personalized Real-to-Sim-to-Real Navigation with Gaussian Splats from a Mobile Device*. arXiv:**2509.17430**. Navigation; SRCC 0.87–0.97 printed as a range, exact values inside its Figure 1.
- **MolmoSpaces** — *MolmoSpaces: A Large-Scale Open Ecosystem for Robot Navigation and Manipulation*. arXiv:**2602.11337**. Prints CIs on its correlations inside its figure; text and figure disagree on the pick-task Spearman (0.98 vs 1.00).

- **Mem-World** — *Mem-World: Memory-Augmented Action-Conditioned World Models for Persistent Robot Manipulation*. arXiv:**2606.18960**. $r = 0.97$, $p < 0.001$ over two sibling finetunes; §8.

Added by the pre-submission rerun (2026-07-21; §8.1, Scope)

- **Colosseum V2** — *Colosseum V2: Benchmarking Generalization for Vision Language Action Models*. arXiv:**2605.27759**. Avg $R^2 = 0.798$ and avg Spearman = 0.916 across perturbation conditions of one ACT policy on three real tasks; $k = 1$ (§8.1).

Excluded from the counts (§8.1, Rule 3)

- **GSWorld** — *GSWorld: Closed-Loop Photo-Realistic Simulation Suite for Robotic Manipulation*. arXiv:**2510.20813**. Claims a “strong correlation”; computes no coefficient.
- **RoboSimGS** — *High-Fidelity Simulated Data Generation for Real-World Zero-Shot Robotic Manipulation Learning with Gaussian Splatting*. arXiv:**2510.10637**. Reports no sim-vs-real correlation.
- **Interactive World Simulator** — *Interactive World Simulator for Robot Policy Training and Evaluation*. arXiv:**2603.08546**. Plots a sim-vs-real scatter (4 policies); prints no coefficient.
- **AutoEval** — *AutoEval: Autonomous Evaluation of Generalist Robot Manipulation Policies in the Real World*. arXiv:**2503.24278**. Its headline Pearson 0.942 is real-to-real (autonomous vs human evaluation); its SIMPLER-vs-real correlations are plotted per task with no numeric coefficient.
- **What Are We Actually Benchmarking in Robot Manipulation?** — arXiv:**2606.04233**. A benchmark-protocol audit rather than a sim-real correlation study; finds only 19.8% of LIBERO and 19.7% of SimplerEnv SOTA claims provably significant. Closest related work; see §1.1.
- **GigaWorld-1** — *GigaWorld-1: A Roadmap to Build World Models for Robot Policy Evaluation*. arXiv:**2607.02642**. Defines the real-vs-predicted correlation (its Eq. 4) as its central quantity but prints no value for it; screened 2026-07-21.

Methodological precedent

- Henderson et al., *Deep Reinforcement Learning that Matters*. arXiv:**1709.06560**.
- Agarwal et al., *Deep Reinforcement Learning at the Edge of the Statistical Precipice (rliable)*. arXiv:**2108.13264**.

Statistical lineage (§1.1)

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